

Sreeaswin R

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ABOUT ME

Systems programmer with hands-on experience building low-level networked applications in **C** and **C++**, including custom memory management and concurrent server design. Strong foundation in data structures, algorithms, and operating system concepts, complemented by full-stack and AI/ML project exposure. Driven to build performant, maintainable, and production-grade systems.

EDUCATION

Amrita Vishwa Vidyapeetham

*Bachelor of Technology in **Computer and Communication Engineering** (7.55 CGPA)*

Ettimadai, TN

Aug 2023 – June 2027

AVP National Metric School

Senior High School

Tiruppur, TN

April 2021 – April 2023

PROJECTS

Custom Memory Allocator | *C, Memory Management, Data Structures, OS Concepts*

- Implemented **malloc**, **free**, **calloc**, and **realloc** from scratch in C using **sbrk()** to request heap memory directly from the OS.
- Designed a **linked-list block structure** with metadata headers tracking block size and allocation state, mirroring real allocator internals.
- Applied **first-fit search** and **free-block coalescing** to minimise heap fragmentation across repeated allocation cycles.
- Reinforced understanding of **heap memory layout**, **pointer arithmetic**, and **low-level memory management** as studied in OS coursework.

Distributed Key-Value Store (Redis from Scratch) | *Go, TCP, In-Memory Storage, Concurrency*

- Built a Redis-like in-memory key-value store from scratch, supporting **SET**, **GET**, **DEL** commands over a custom TCP protocol.
- Implemented a **concurrent TCP server** using goroutines and channels, applying OS-level concurrency and process scheduling concepts.
- Designed in-memory data structures with efficient access patterns, demonstrating practical **DSA** and memory management principles.
- Strengthened understanding of **low-level networking**, **concurrency control**, and **system design** analogous to C/C++ systems programming.

Distributed Load Balancer | *Go, TCP, Networking, Concurrency*

- Built a load balancer in Go to distribute incoming client requests across multiple backend servers using round-robin routing.
- Implemented **concurrent request handling** with goroutines, applying process management and inter-process communication concepts from OS.
- Explored **fault tolerance**, **request routing**, and **backend scalability** under concurrent load conditions.
- Analysed system bottlenecks and traffic distribution behaviour, developing a **problem-solving and troubleshooting** mindset.

AgriSupply – Fertilizer Management System | *Node.js, Express.js, MongoDB, React, Tailwind CSS*

- Developed a full-stack inventory management system covering the complete **software development lifecycle (SDLC)** from design to deployment.
- Implemented **role-based workflows** for suppliers and administrators using Express.js and MongoDB, applying DBMS concepts in production.
- Built a responsive frontend with React and Tailwind CSS; improved data visibility and operational efficiency across the supply chain.

RELEVANT COURSEWORK

Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Machine Learning

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Go
Technologies: MySQL, MongoDB
Frameworks and Libraries: React, Node.js, Next.js, Tailwind CSS, Pandas, NumPy, Matplotlib
Developer Tools: Git, Docker, GitHub Actions, Postman, Linux

EXTRACURRICULAR ACTIVITIES

President, Intel IoT Club: Led hackathons and workshops in AI/IoT, engaging **400+ students**.
Winner, Random Compile Coding Contest: Amrita Vishwa Vidyapeetham, Chennai.
Oracle Cloud Certified: AI Foundations Associate (2024), Foundations Associate (2024).